

Time Is of the Essence

Experimental Evidence on the Effects of Job Application Timing

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October 3, 2026

Timing in the Labor Market

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- **This paper:** how firms' timing decisions shape who gets considered

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- What is the **cost of speed** for firms?

A stopping model quantifies the tradeoff: **6.6%** relative loss in match quality

Literature Contribution

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- **This paper: firm-side timing of contact initiation**

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■ Audit-study design

Gaddis (2018), Kessler et al. (2019), and Neumark (2018)

- This paper: multiple applications as tracers

Early Application Advantage

Experiment Design (Kline et al., 2022)

vacancy open



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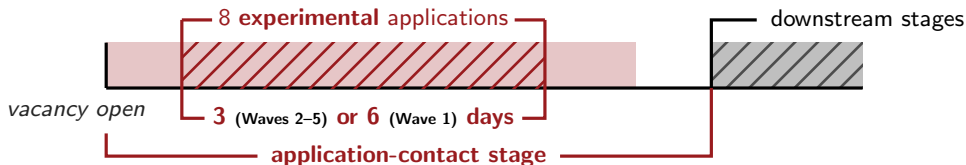
- **11,114** entry-level vacancies at 108 large U.S. firms

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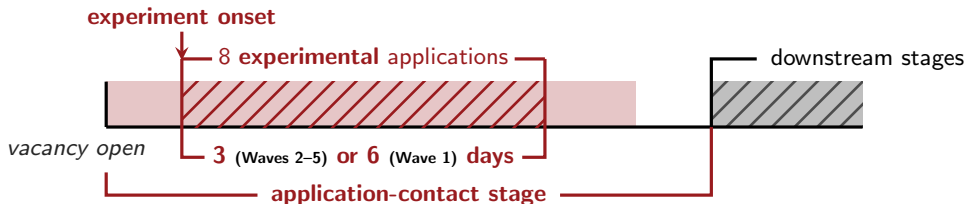
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- **outcome**: employer contact within 30 days (~24%)

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- **8 randomized applications** in 4 pairs over 3 (Waves 2-5) or 6 days (Wave 1)
- **outcome**: employer contact within 30 days (~24%)
- **Market conditions** measured at the first experimental submission

► covariates

Submission Order and Sample Completeness

Order	Full Sample	<i>Incomplete</i>
1	11,114	1,285
2	11,031	1,202
3	10,682	853
4	10,606	777
5	10,278	449
6	10,204	375
7	9,899	70
8	9,829	0
<i>N</i> applications	83,643	5,011
<i>N</i> vacancies	11,114	1,285
<i>N</i> firms	108	39

Submission Order and Sample Completeness

Order	Full Sample	<i>Incomplete</i>	Wave 1	Waves 2–5
1	11,114	1,285	2,063	9,051
2	11,031	1,202	2,040	8,991
3	10,682	853	1,883	8,799
4	10,606	777	1,879	8,727
5	10,278	449	1,748	8,530
6	10,204	375	1,734	8,470
7	9,899	70	1,604	8,295
8	9,829	0	1,590	8,239
<i>N</i> applications	83,643	5,011	14,541	69,102
<i>N</i> vacancies	11,114	1,285	2,063	9,051
<i>N</i> firms	108	39	85	105

Measuring Application Timing: Early Indicator

Define the first 4 submissions (out of 8) to a job as **early**:

$early_i = \mathbf{1}$ (application i among the first 4 applications)

41,112 submissions (49.15%) are classified as early.

Measuring Application Timing: Early Indicator

$early_i = \mathbf{1}$ (application i among the first 4 applications)

Estimate

$$Y_i = \alpha_0 + \beta \cdot early_i + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

where

- $\mathbf{X}_i$ includes the randomized resume characteristics, $\eta_{j(i)}$ is a vacancy fixed effect.
- Standard errors are clustered at the vacancy level (Abadie et al., 2023).

Measuring Application Timing: Early Indicator

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Estimate

$$Y_i = \alpha_0 + \beta \cdot early_i + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

$early_i$	$Black_i$	$Female_i$
0.0116***	-0.0205***	-0.0021
(0.0024)	(0.0018)	(0.0022)

Full sample: $N = 83,643$

► robustness

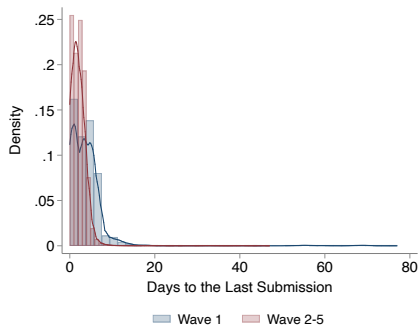
► by order

Measuring Application Timing: Days to Last Submission

Let $D2L_i$ be the number of days between application i and the **final experimental submission** to the same vacancy,

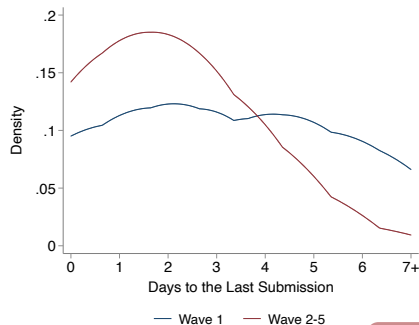
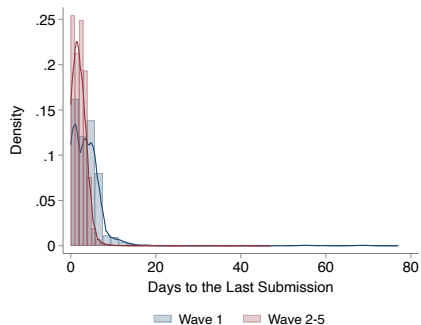
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Measuring Application Timing: Days to Last Submission

Let $D2L_i$ be the number of days between application i and the **final experimental submission** to the same vacancy, define $D2L_i^t = \min\{D2L_i, 7\}$.



► by order

Early Application Advantage: Days to Last Submission

Estimate

$$Y_i = \alpha_0 + \delta \cdot D2L_i^t + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

(1)	
Full Sample	
$D2L_i^t$	0.0040*** (0.0008)
N	83,643

► robustness

Early Application Advantage: Days to Last Submission

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$$Y_i = \alpha_0 + \delta \cdot D2L_i^t + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

	(1) Full Sample	(2) Wave 1	(3) Waves 2–5
$D2L_i^t$	0.0040*** (0.0008)	0.0036*** (0.0013)	0.0043*** (0.0010)
N	83,643	14,541	69,102

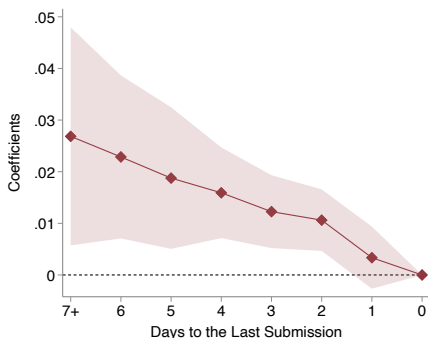
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Early Application Advantage: Days to Last Submission

$$Y_i = \alpha + \sum_{d=1}^7 \delta_d \mathbf{1}\{D2L_i^t = d\} + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

Early Application Advantage: Days to Last Submission

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► robustness

► heterogeneity

Mechanism: Contact Initiation

A Firm-Side Mechanism



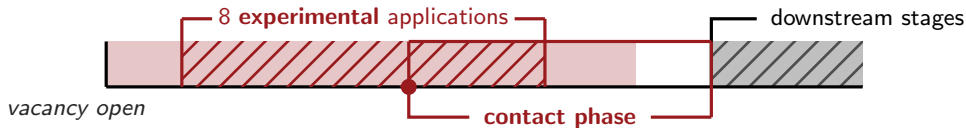
- **So far:** Earlier experimental applications are more likely to be contacted

A Firm-Side Mechanism



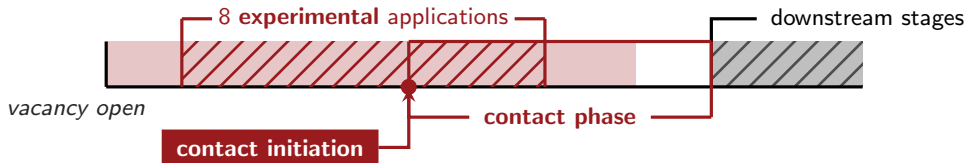
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- **Possible families of explanations:** **direct-timing** versus **pool-based**

A Firm-Side Mechanism



- **So far:** Earlier experimental applications are more likely to be contacted
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- **Hypothesis:** Applications arriving in the **contact phase** face a different recruiting environment

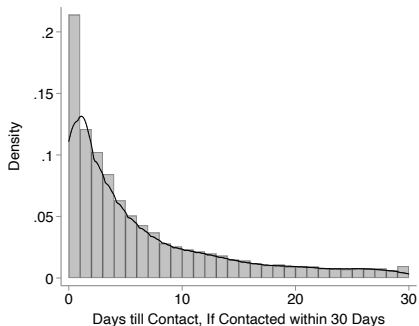
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- **So far:** Earlier experimental applications are more likely to be contacted
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- **Hypothesis:** Applications arriving in the **contact phase** face a different recruiting environment
- **Key state change:** the timing of **contact initiation**

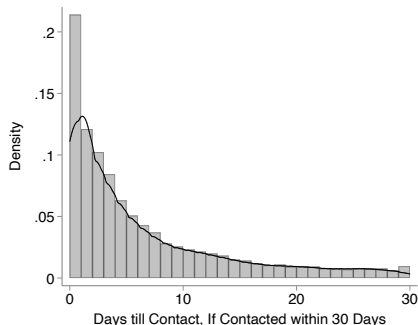
Recruiters Initiate Contact Early

Applications: Contacts happen fast

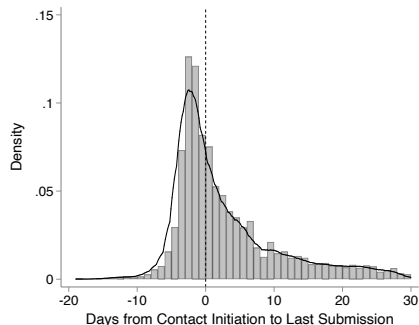


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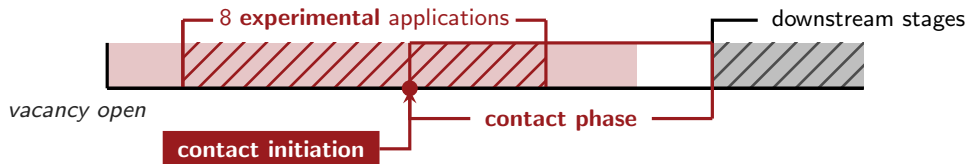
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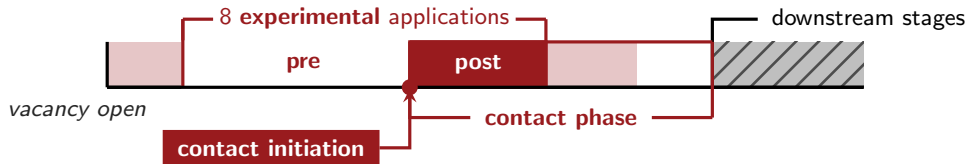
Vacancies: Contact initiation is early



Contact Initiation: Changes in the Recruiting Environment



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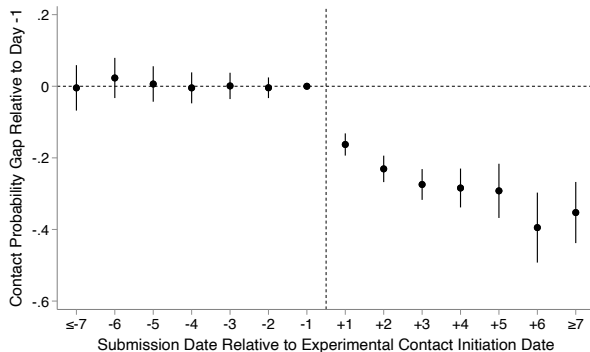


Estimate:

$$Y_i = \sum_{k \in \mathcal{K}, k \neq -1} \beta_k \cdot \mathbf{1}(e_{j(i)} = k) + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i,$$

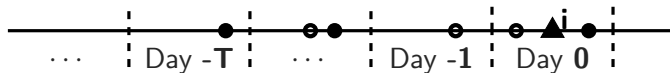
where $e_{j(i)} = \text{submission date}_i - \text{contact initiation date}_{j(i)}$, $\mathbf{X}_i$ includes randomized resume controls, submission-order FEs, and submission-day-of-week FEs.

Contact Initiation: Changes in the Recruiting Environment

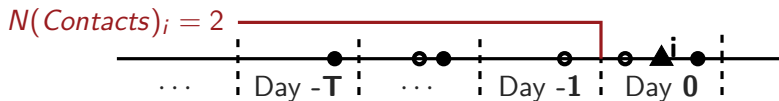


Contact probability drops sharply **after contact initiation**

Constructing the Post-Initiation Indicator

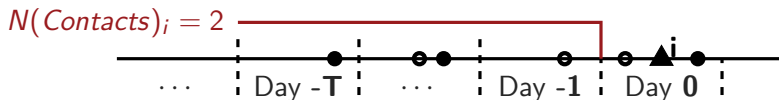


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	$N(Contacts)_i = 0$	$N(Contacts)_i > 0$	Diff.
	.2641*** (.0034)	.0974*** (.0125)	-.1667*** (.0127)
N	70,293	8,339	78,632

Contact probability decreases by **63%** after contact initiation.

► $N(Contacts)_i^{alt}$

Contact Initiation Absorbs the Timing Advantage

$$Y_i = \alpha_0 + \beta \cdot \text{timing}_i + \iota \cdot \mathbf{1}(N(\text{Contacts})_i > 0) + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

	(1)	(2)	(3)	(4)
<i>early_i</i>	0.0112*** (0.0024)	-0.0099*** (0.0022)		
<i>D2L_i^t</i>			0.0038*** (0.0008)	-0.0039*** (0.0007)
$\mathbf{1}(N(\text{Contacts})_i > 0)$		-0.1733*** (0.0086)		-0.1760*** (0.0087)
<i>N</i>		78,632		

► $N(\text{Contacts})_i^{\text{alt}}$

Contact Initiation Absorbs the Timing Advantage

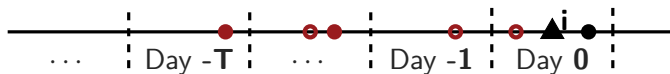
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But contact initiation is **endogenous**.

► $N(\text{Contacts})_i^{\text{alt}}$

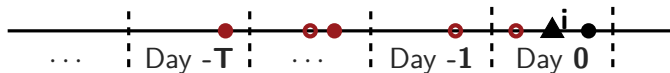
Instrumenting Contact Initiation



- **Intuition:** the quality of applications before i is **randomized**

[▶ model details](#)[▶ \$\hat{p}_{ij}\$ validation](#)

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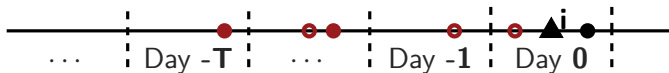


- **Intuition:** the quality of applications before i is **randomized**
- **Quality proxy:** train prediction models on the **first experimental application** to each vacancy

$$\hat{p}_{ij} = \Pr(\text{contact}_{ij} = 1 \mid \mathbf{x}_{ij}, \mathbf{z}_j)$$

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- **Instrument:** best out-of-fold predicted screening value **at least one day before i**

$$Q_i^{\text{best}} = \max_{k: \text{day}(k) < \text{day}(i)} \{\hat{p}_{kj}\} \rightarrow \mathbf{1}(N(\text{Contacts})_i > 0)$$

► model details

► $\hat{p}_{ij}$ validation

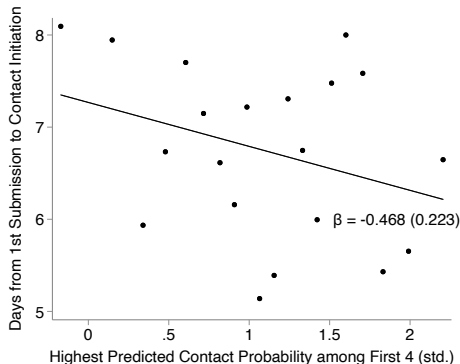
IV Estimation: Contact Initiation and Subsequent Contacts

$Q_i^{best} \xrightarrow{\text{first stage}} \mathbf{1}(N(\text{Contacts})_i > 0) \xrightarrow{\text{second stage}} \text{contact}$

	All Vacancies		Contacted Vacancies	
	(1)	(2)	(3)	(4)
	IV-2SLS	OLS	IV-2SLS	OLS
$\mathbf{1}(N(\text{Contacts})_i > 0)$	-1.2642* (0.7380)	-0.1204*** (0.0206)	-1.1077* (0.5950)	-0.1275*** (0.0217)
First Stage	0.0078** (0.0039)		0.0235** (0.0098)	
Kleibergen-Paap F	5.218		7.602	
Anderson-Rubin p	0.025		0.023	
N	39,300		16,142 (4,036 vacancies)	

► robustness

Best-So-Far Application as Contact Initiation Trigger



Higher Early Screening Value, **Earlier** Contact Initiation

The Evidence Points to Threshold-Based Screening

The evidence supports a threshold-based mechanism:

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▶ A

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▶ A

▶ B

▶ C

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- **Pure batching:** Later submissions are generally **contacted later** ▶ D

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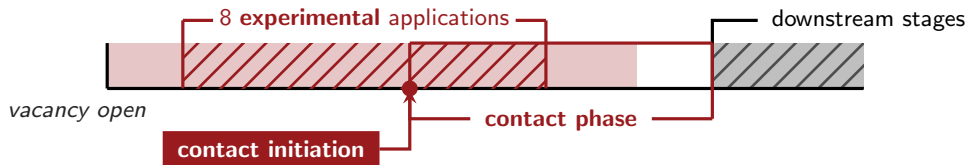
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- **Full-pool comparison**: Screening value of later applications **does not predict earlier contacts** ▶ C
- **Pure batching**: Later submissions are generally **contacted later** ▶ D

The key firm choice: Initiate contact when the best so far **crosses a threshold**

A Model of Contact Initiation

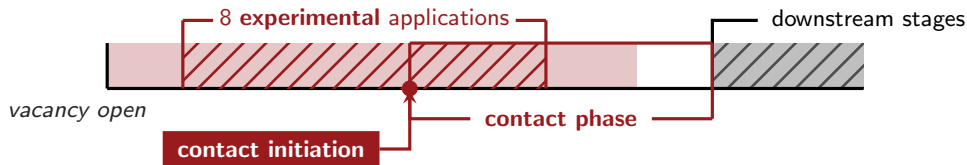
Application Timing in the Labor Market



So far:

- significant early application advantages
- driven by recruiters initiating contacts before the application pool is complete
- contact initiation triggered by best-so-far passing a threshold

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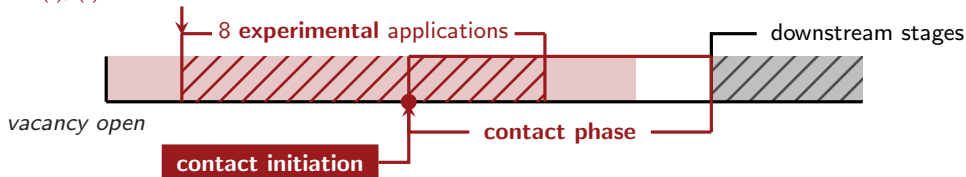
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How does the timing of contact initiation vary with **labor-market conditions**?

Application Timing in the Labor Market

$Z_{m(i),t(i)}$ at experiment onset



$$Y_i = \alpha_0 + \delta D2F_i^t + \phi (D2F_i^t \times Z_{m(i),t(i)}) \\ + \phi^{\text{pre}} (D2F_i^t \times \bar{Z}_{m(i),t(i)-3:t(i)-1}) + \gamma \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i,$$

$Z_{m(i),t(i)}$: conditions in local labor market $m(i)$ in the month of the **first submission** $t(i)$

Contact Timing by Labor Market Tightness

Dep. Var.	Contacted	Contact Initiation
$D2F_i^t$	-.0026*** (.0010) [.0010]	-0.0135*** (.0006) [.0008]
$D2F_i^t \times \log(V/U_{m(i),t(i)})$	-0.0078 (.0057) [.0056]	-0.0291*** (.0040) [.0049]
N	78,632	78,632

▸ $D2F_i^t$

▸ JOLTS

Contact Timing by Labor Market Tightness

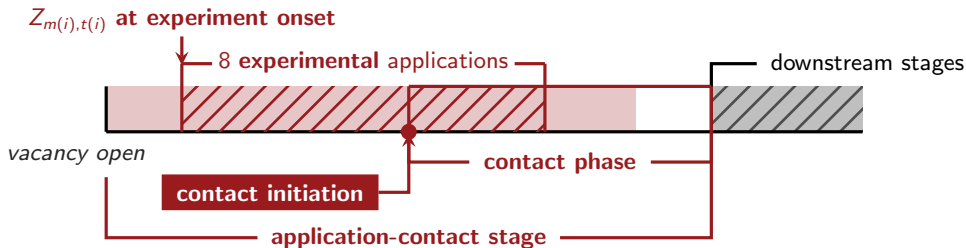
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more vacancies to fill $\Rightarrow$ firms move faster $\Rightarrow$ early advantage $\uparrow$

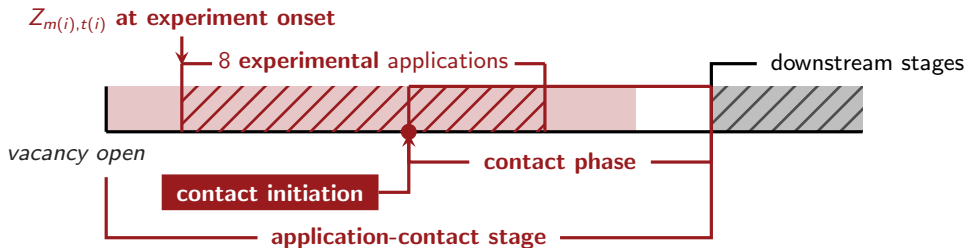
► $D2F_i^t$

► JOLTS

Modeling the Application-Contact Stage



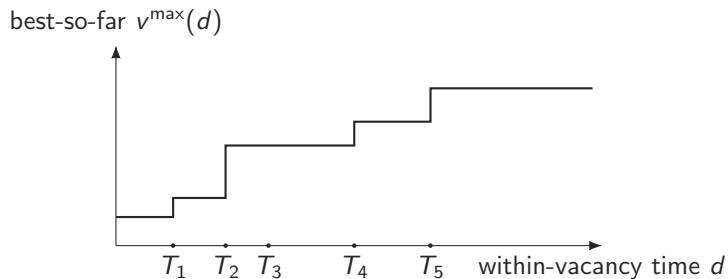
Modeling the Application-Contact Stage



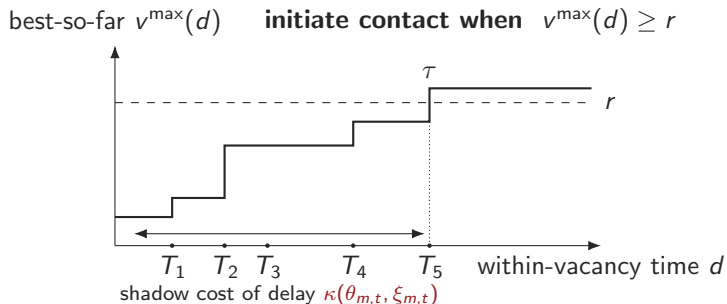
Model: a partial-equilibrium firm-side policy rule that

- quantifies the **recruiting-speed** versus **screening-value tradeoff** of firms
- accounts for **organic application inflows**
- separates two underlying economic forces: **arrival intensity** and **the cost of delay**

The Threshold Rule and Firms' Tradeoff



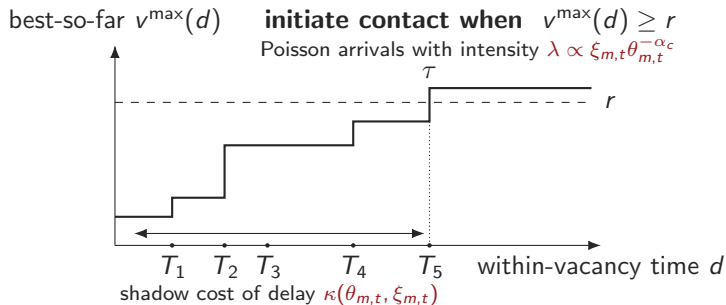
The Threshold Rule and Firms' Tradeoff



The firm's optimization problem is

$$\max_{\tau} \mathbb{E} \left[\underbrace{e^{-\delta\tau} v^{\max}(\tau)}_{\text{value of the best application}} - \underbrace{\int_0^{\tau} e^{-\delta s} \kappa ds}_{\text{cost of delay}} \right]$$

The Threshold Rule and Firms' Tradeoff



The optimal threshold is characterized by

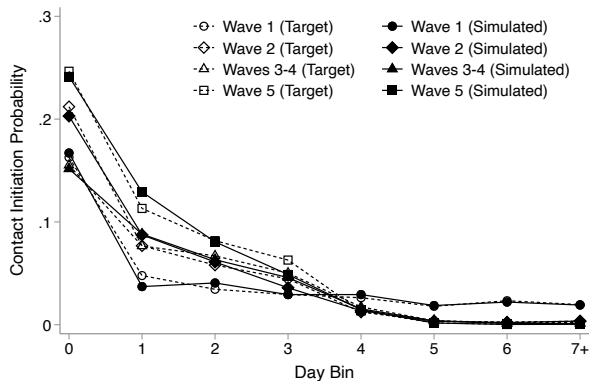
$$\underbrace{\delta r + \kappa}_{\text{cost of delay}} = \underbrace{\lambda \mathbb{E}[(v - r)^+]}_{\text{option value of another applicant}}, \quad v \sim \mathcal{N}(0, 1)$$

Identification and Estimation

- **Observed**: submission dates and experimental contact initiators
- **Mapping**: threshold crossing and survival from earlier initiation
- **Estimation**: Method of Simulated Moments
- **Two families of moments**: 33 moments, 9 or 12 parameters
 - Experimental initiator mass by day ($D2F_i^t = 0, \dots, 6$), and residual *none* mass
 - tightness gradient of contact initiation
- Contact-initiation policy on a **normalized screening-value** scale

► targeted moments

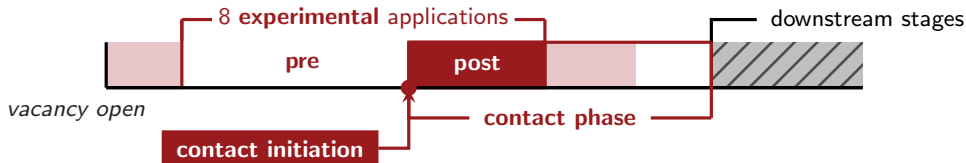
Identification and Estimation



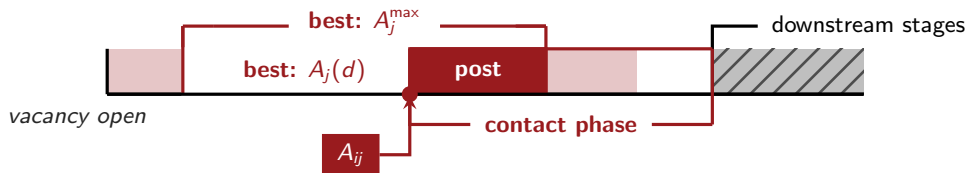
► fit summary

► robustness

The Screening-Value Cost of Early Contact Initiation



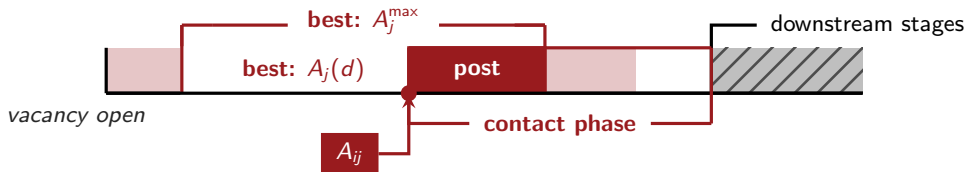
The Screening-Value Cost of Early Contact Initiation



If initiation is assigned to experimental application i , then

$$A_j^{\max} - A_{ij} = \underbrace{A_j^{\max} - A_j(d)}_{\text{timing component}} + \underbrace{A_j(d) - A_{ij}}_{\text{within-pool value gap}}$$

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Construct two *conditional* loss measures:

$$\text{Missed-Best Probability} = \Pr(A_j(d) < A_j^{\max}), \quad \text{Relative Timing Loss} = \frac{\mathbb{E}[A_j^{\max} - A_j(d)]}{\mathbb{E}[A_{ij}]}$$

Early Contact Initiation Has Meaningful Screening-Value Costs

	$\bar{P}^{\text{exp}}(\hat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	0.4079 [0.3985, 0.4177]	0.5156 [0.5069, 0.5234]	0.0656 [0.0642, 0.0674]

► robustness

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Baseline	0.4079 [0.3985, 0.4177]	0.5156 [0.5069, 0.5234]	0.0656 [0.0642, 0.0674]
One Day Earlier Δ		0.1175 [0.1144, 0.1209]	0.0217 [0.0211, 0.0223]
One Day Later Δ		-0.1713 [-0.1733, -0.1691]	-0.0292 [-0.0301, -0.0285]

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Early Contact Initiation Has Meaningful Screening-Value Costs

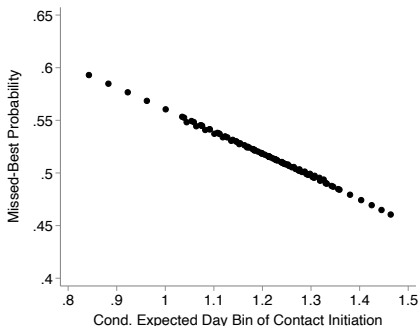
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$$\text{Wage-equivalent timing loss} = \underbrace{\text{RTL}}_{0.0656} \times \underbrace{\bar{w}}_{\$47,150} \times \underbrace{\Delta p_{\text{contact}}}_{63.12\%} \times \underbrace{\pi_{H|C}}_{[0.20, 0.40]} = [\$390, \$781]$$

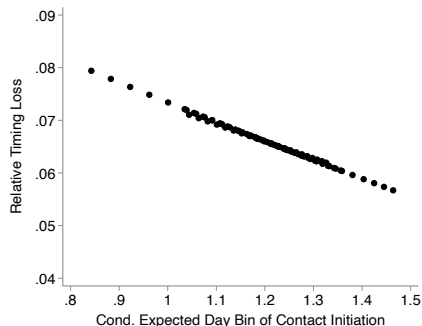
► robustness

Firms' Tradeoff: Speed-Loss Frontier

Across recruiting environments, earlier contact comes with larger timing losses:



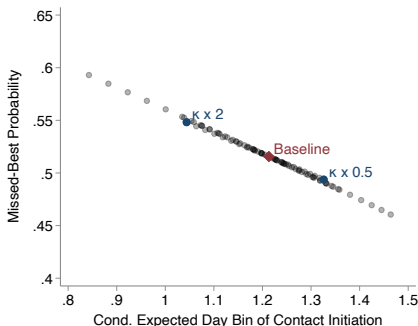
Missed-Best Probability



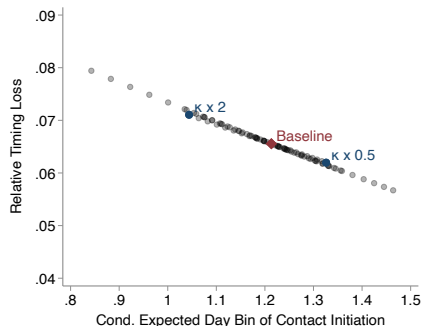
Relative Timing Loss

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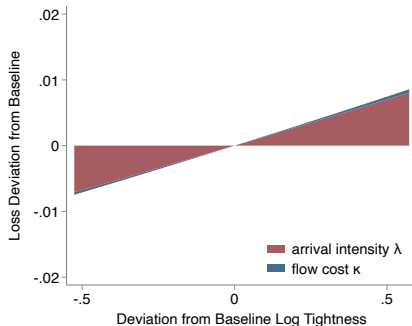


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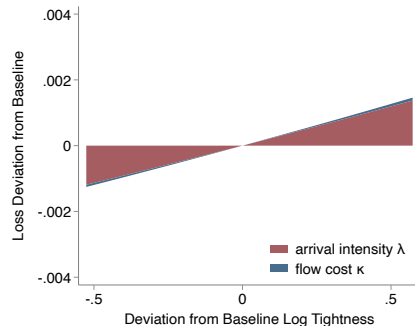
► magnitude

Quantifying Firms' Tradeoff: Counterfactuals

Tighter markets increase timing losses, mainly through the **arrival-intensity channel**:



Missed-Best Probability

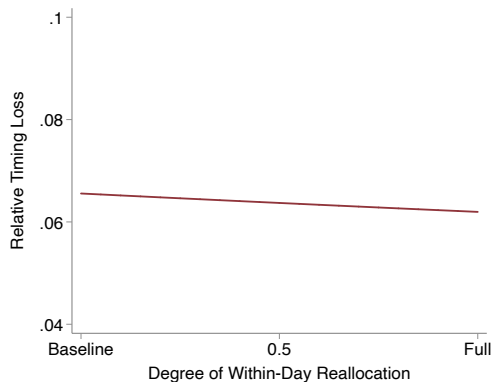


Relative Timing Loss

► other counterfactuals

Quantifying Firms' Tradeoff: Counterfactuals

Within-pool reallocation **cannot eliminate timing loss:**



Welfare Implications of Timing Frictions

Sequential arrival creates a queue-position wedge **even when firms optimize**:

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- **micro-to-macro link:** recruiting conditions shape when firms act and who receives consideration
 - **time-varying matching efficiency** (Barlevy et al., 2024; Barnichon and Figura, 2015; Davis, Faberman, et al., 2006; Sedláček, 2014)
 - **cyclical match quality** (Barlevy, 2002; Bellou and Kaymak, 2021; Bils et al., 2023)

Conclusion

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- **Future work** can
 - embed the application-timing wedge in a **general equilibrium** model
 - study how recruiters **update their beliefs**

Thank you!

Contact me: saizhang@usc.edu

Market Conditions and Other Covariates

- **Main measure at experiment onset:** state-month labor-market tightness
 - job openings per unemployed worker (JOLTS and LAUS)
- **Additional recruiting conditions**
 - hires, quits, layoffs and discharges, and job-opening rates (JOLTS)
- **Descriptive external-validity covariates**
 - industry and job content (O*NET; BLS CES and ECI)
 - state income, institutions, and employment composition (IDDA, Kondo et al., 2023) (DOL; LAUS)
 - firm organization, financial characteristics, and job postings (Compustat; Revelio Labs)

Construction of Application Timing Measures

	$early_i$	$early_i^2$	$early_i^3$	$Order_i$	$D2L_i^t$	$N(Contacts)_i$
$early_i$	1					
$early_i^2$	0.9786	1				
$early_i^3$	0.9569	0.9710	1			
$Order_i$	-0.8164	-0.8076	-0.8006	1		
$D2L_i^t$	0.7169	0.7195	0.7146	-0.7406	1	
$N(Contacts)_i$	-0.2048	-0.2006	-0.1993	0.2385	-0.1790	1

- $early_i = 1$ (application i among the first 4 applications)
- $early_i^2 = 1$ (application i not among the last 4 applications)
- $early_i^3 = 1$ (application i not among the last 2 pairs of applications)

Balance of Timing Measures

Dep. Var.	(1) <i>early</i> ¹	(2) <i>early</i> ²	(3) <i>early</i> ³	(4) <i>Order</i>	(5) <i>Order</i> ^{pair}	(6) <i>D2L</i> ^t
Balanced	1.55 [0.1228]	1.37 [0.1938]	1.19 [0.2967]	0.73 [0.6864]	1.02 [0.4212]	1.25 [0.2568]
Full	1.07 [0.3831]	0.98 [0.4564]	0.83 [0.5857]	0.63 [0.7692]	0.83 [0.5876]	0.97 [0.4618]
Complete	0.99 [0.4481]	0.99 [0.4481]	0.82 [0.5988]	0.53 [0.8540]	1.11 [0.3519]	0.92 [0.5040]
Incomplete	1.10 [0.3566]	1.06 [0.3931]	0.87 [0.5478]	0.93 [0.5007]	1.03 [0.4142]	0.75 [0.6610]

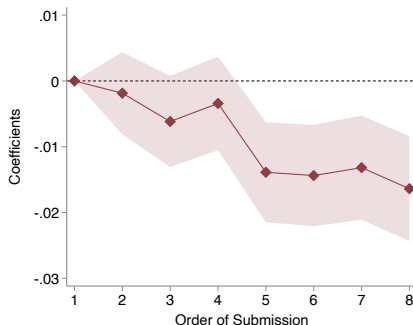
Robustness: Early Indicator

	(1) Complete	(2) Incomplete	(3) Wave 1	(4) Waves 2–5	(5) Full Sample	(6) Full Sample
$early_i$	0.0112*** (0.0024)	0.0240* (0.0127)	0.0128** (0.0061)	0.0113*** (0.0025)		
$early_i^2$					0.0135*** (0.0023)	
$early_i^3$						0.0105*** (0.0023)
N	78,632	5,011	14,541	69,102	83,643	83,643

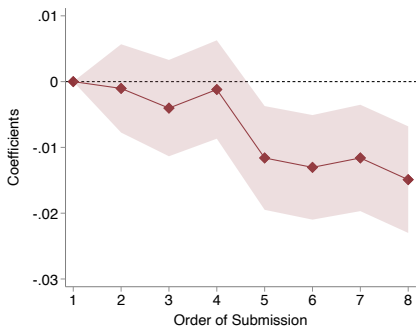
[► construction](#)
[► back](#)

Early Application Advantage: Submission Order

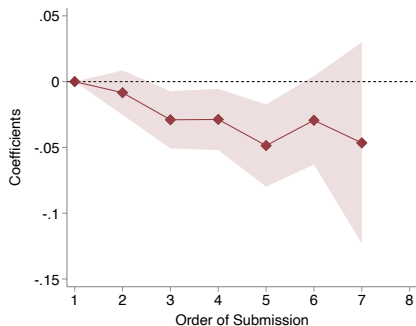
$$Y_i = \alpha + \sum_{k=2}^8 \beta_k \mathbf{1}\{\text{Order}_i = k\} + \gamma' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

[▶ robustness](#)[▶ back](#)

Robustness: Submission Order



Complete vacancies



Incomplete vacancies

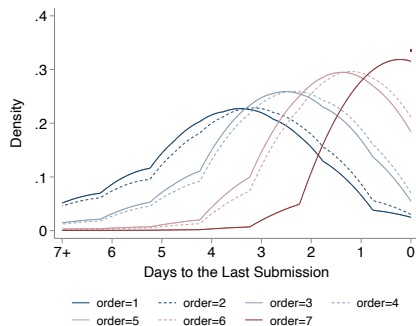
► back

Robustness: Days to Last Submission

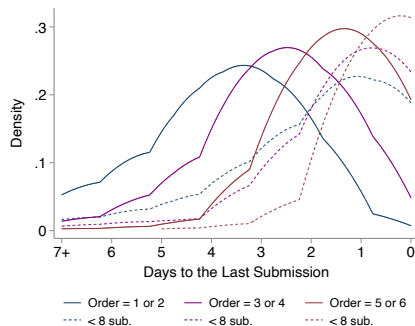
	(1) Full Sample	(2) Wave 1	(3) Waves 2–5	(4) Complete	(5) Incomplete
$D2L_i$ (w.)	0.0039*** (0.0008)	0.0034*** (0.0013)	0.0042*** (0.0010)		
$D2L_i^t$				0.0038*** (0.0008)	0.0087** (0.0035)
N	83,643	14,541	69,102	78,632	5,011

[▶ back](#)

Robustness: Days to Last Submission



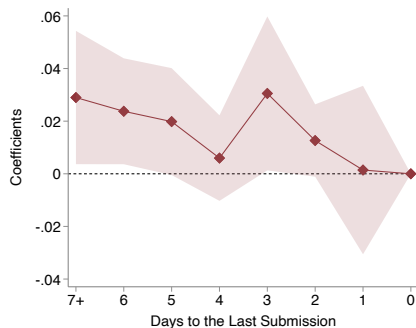
Full Sample



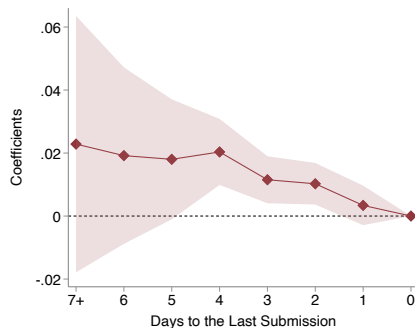
Complete vs. Incomplete

► back

Robustness: Days to Last Submission



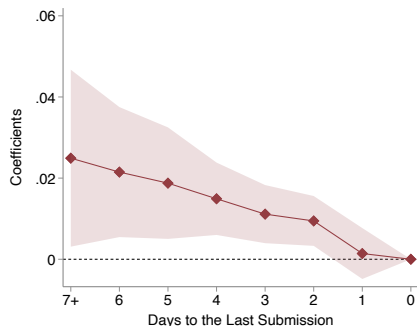
Wave 1



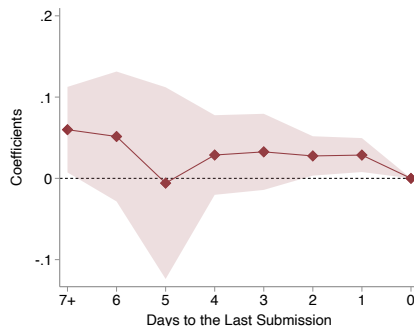
Waves 2-5

▶ back

Robustness: Days to Last Submission



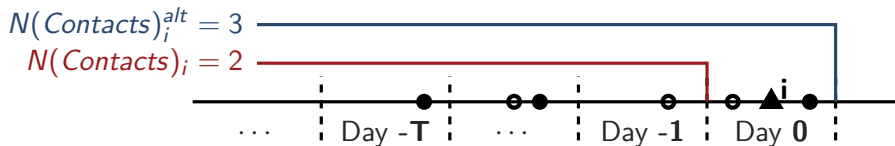
Complete



Incomplete

► back

Constructing the Post-Initiation Indicator



	$N(\text{Contacts})_i^{alt} = 0$	$N(\text{Contacts})_i^{alt} > 0$	Diff.
	.2686*** (.0035)	.1536*** (.0116)	-.1150*** (.0123)
<i>N</i>	67,082	11,550	78,632

Contact Initiation Absorbs the Timing Advantage

	(1)	(2)	(3)	(4)
$early_i$	0.0112*** (0.0024)	-0.0017 (0.0022)		
$D2L_i^t$			0.0038*** (0.0008)	-0.0011 (0.0008)
$1(N(Contacts)_i^{alt} > 0)$		-0.1257*** (0.0094)		-0.1273*** (0.0094)
N		78,632		

[▶ back](#)

Machine Learning Prediction Models

- **Training sample:** first experimental application to each vacancy, $N = 11,114$
- **Input predictors:** all resume characteristics, SOC3 categories of vacancies, detailed resume entry categories (1,512 job-type categories, 23 job titles, duration in months)
- **Out-of-fold construction:**
 - Posting-level five-fold cross-fitting
 - Each vacancy scored using a model trained on other vacancies
- **Model candidates:** Elastic-net logistic regression, ExtraTrees classifier, CatBoost gradient boosting (with native and encoded category handling)
- **Model selection criteria:** prediction accuracy and within-vacancy score dispersion

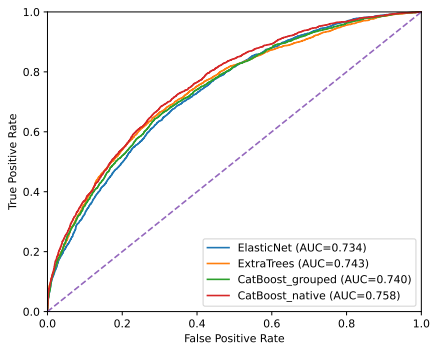
[▶ performance score](#)[▶ visual comparison](#)[▶ back](#)

Pre-Tuning Model Comparison

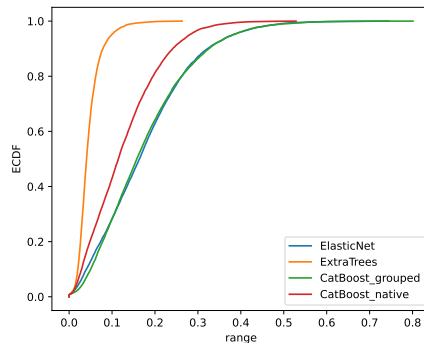
Model	ROC-AUC	Log Loss	Brier	Average Precision
CatBoost (native categories)	0.758	0.474	0.154	0.525
ExtraTrees (encoded)	0.743	0.491	0.160	0.511
CatBoost (encoded)	0.740	0.484	0.158	0.504
Elastic-net logit (encoded)	0.734	0.491	0.161	0.488

[► model details](#)[► back](#)

Machine Learning Model Comparison



ROC Curve



Predicted Score Range

► model details

► back

Prediction Score Validation

	(1)	(2)	(3)	(4)	(5)	(6)
		All Vacancies		Vacancies w. ≥ 1 Contact		
Main index	0.1587*** (0.0404)			0.2807*** (0.0715)		
Alternative 1		0.1240*** (0.0282)			0.2525*** (0.0573)	
Alternative 2			0.1078*** (0.0278)			0.2060*** (0.0533)
Vacancy FEs	✓	✓	✓	✓	✓	✓
<i>N</i>	78,632	78,632	78,632	32,288	32,288	32,288

[▶ back](#)

Prediction Score Validation

	(1)	(2)	(3)	(4)	(5)	(6)
		All Vacancies		Vacancies w. ≥ 1 Contact		
Main index	1.0120*** (0.0182)			0.6084*** (0.0253)		
Alternative 1		1.1650*** (0.0214)			0.6988*** (0.0293)	
Alternative 2			1.1340*** (0.0208)			0.6743*** (0.0285)
St-Mth FEs	✓	✓	✓	✓	✓	✓
<i>N</i>	78,632	78,632	78,632	32,288	32,288	32,288

[▶ back](#)

Prediction Score Validation

Dep.Var.	(1) Main Index	(2)	(3) Alternative 1	(4)	(5) Alternative 2	(6)
$early_i^1$	-0.0003 (0.0002)		0.0004 (0.0003)		0.0003 (0.0003)	
$D2L_i^t$		-0.0001 (0.0001)		0.0002 (0.0001)		0.0001 (0.0001)
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	78,632	78,632	78,632

[▶ back](#)

IV Estimation: Robustness

Using an alternative index built from a CatBoost model with frequency-based grouping

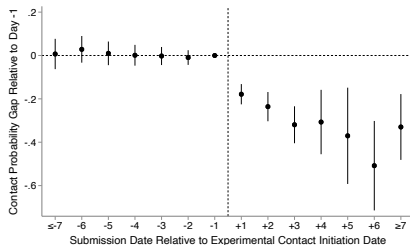
	All Vacancies		Contacted Vacancies	
	(1) IV-2SLS	(2) OLS	(3) IV-2SLS	(4) OLS
$\mathbf{1}(N(\text{Contacts})_i > 0)$	-0.6520 (0.3981)	-0.1205*** (0.0206)	-0.5742* (0.3469)	-0.1277*** (0.0217)
Kleibergen-Paap F	10.781		15.172	
Anderson-Rubin p	0.083		0.085	
N	39,300		16,142 (4,036 vacancies)	

IV Estimation: Robustness

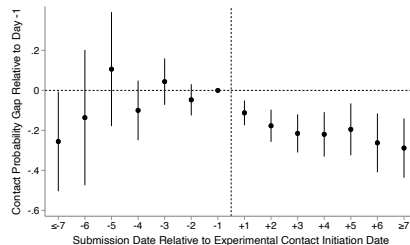
Using an alternative index built from a CatBoost model with text-based grouping

	All Vacancies		Contacted Vacancies	
	(1)	(2)	(3)	(4)
	IV-2SLS	OLS	IV-2SLS	OLS
$1(N(\text{Contacts})_i > 0)$	-1.0857 (0.8640)	-0.1204*** (0.0206)	-0.9264 (0.6883)	-0.1275*** (0.0217)
Kleibergen-Paap F	3.406		5.112	
Anderson-Rubin p	0.112		0.113	
N	39,300		16,142 (4,036 vacancies)	

Contact Initiation and the Best Application Arrival



Before Initiation



After Initiation

► main event study

► back

Alternative Explanation: Pool-Quality Learning

	(1)	(2)	(3)	(4)
	Outcome: Contact Initiation			
	All Vacancies		Vacancies w. ≥ 1 Contact	
Q_i^{best}	0.0078** (0.0039)		0.0235** (0.0098)	
Q_i^{mean}		0.0040 (0.0079)		0.0134 (0.0180)
N	39,300	39,300	16,142	16,142

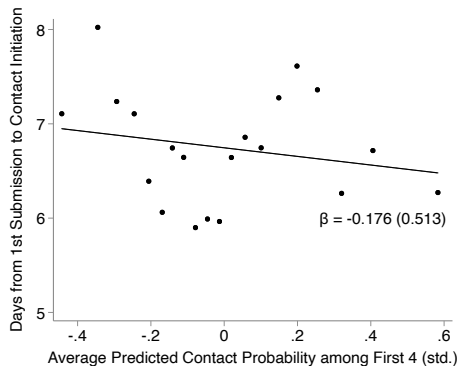
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Alternative Explanation: Pool-Quality Learning

	(1)	(2)	(3)	(4)	(5)	(6)
	All Vacancies		Outcome: Contact			
			Vacancies w. ≥ 1 Contact			
Q_i^{best}	-0.0102** (0.0050)		-0.0099 (0.0064)	-0.0271** (0.0130)		-0.0274* (0.0165)
Q_i^{mean}		-0.0129 (0.0098)	-0.0007 (0.0126)		-0.0309 (0.0233)	0.0008 (0.0295)
N	39,300	39,300	39,300	16,142	16,142	16,142

[▶ back](#)

Alternative Explanation: Pool-Quality Learning



Alternative Explanation: Intrinsic Value of Earliness

	(1) All Vacancies	(2)	(3) Early Contact Init.	(4)	(5) Late Contact Init.	(6)
<i>early</i>	0.0273*** (0.0058)		0.0933*** (0.0076)		-0.0523*** (0.0085)	
<i>D2L^t</i>		0.0096*** (0.0020)		0.0326*** (0.0025)		-0.0243*** (0.0031)
<i>N</i>	32,288	32,288	17,616	17,616	14,672	14,672

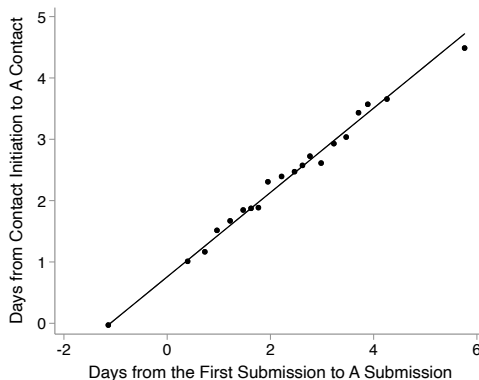
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Alternative Explanation: Full-Pool Comparison

	(1) All Vacancies	(2)	(3) Vacancies w. ≥ 1 Contact	(4)
$Q_i^{best,post}$	0.0053 (0.0048)		0.0122 (0.0118)	
$Q_i^{mean,post}$		0.0027 (0.0099)		0.0044 (0.0233)
N	39,281	39,281	16,143	16,143

[▶ back](#)

Alternative Explanation: Pure Batching



Contact Timing by Labor Market Tightness: Robustness

	(1)	(2)	(3)	(4)	(5)
			Pr (Contact Initiator)		
$D2F_i^t \times \log(\text{hires}_{m(i),t(i)})$	-0.0097* (0.0052)				-0.0054 (0.0063)
$D2F_i^t \times \log(\text{layoffs}_{m(i),t(i)})$		0.0006 (0.0011)			-0.0062*** (0.0014)
$D2F_i^t \times \log(\text{quits}_{m(i),t(i)})$			-0.0166*** (0.0047)		0.0021 (0.0053)
$D2F_i^t \times \log(\text{job opening}_{m(i),t(i)})$				-0.0278*** (0.0055)	0.0024 (0.0103)
$D2F_i^t \times \log(\text{tightness}_{m(i),t(i)})$					-0.0368*** (0.0080)
N	78,632	78,632	78,632	78,632	78,632

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Construction of $D2F_i^t$

$D2F_i^t$ is constructed similarly to $D2L_i^t$:

$$D2F_i^t = \min \{ \text{days between application } i \text{ and the first application}, 7 \}$$

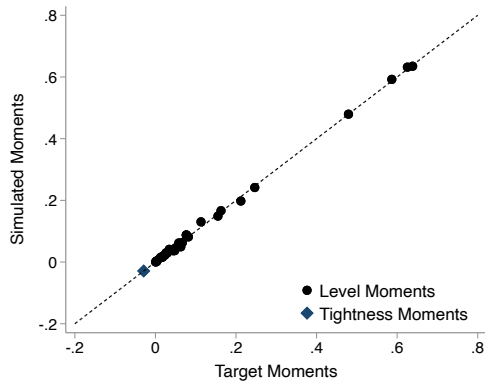
$D2F_i^t$ and $D2L_i^t$ are almost perfectly negatively correlated:

	$D2F_i^t$
$D2L_i^t$	-0.9748*** (0.0018)
Vacancy FEs	✓
N	78,632

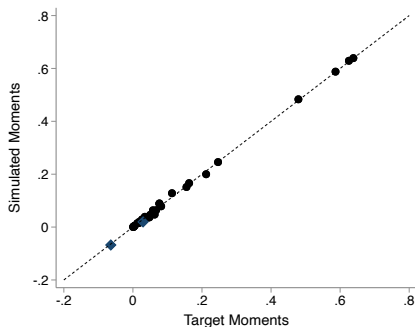
Targeted Moments

<i>Level moments: Contact-initiation day mass</i>								
	$d = 0$	$d = 1$	$d = 2$	$d = 3$	$d = 4$	$d = 5$	$d = 6$	None
$h(j) = 1$	0.1628 (0.0088)	0.0478 (0.0051)	0.0345 (0.0042)	0.0294 (0.0038)	0.0263 (0.0036)	0.0179 (0.0030)	0.0234 (0.0034)	0.6384 (0.0118)
$h(j) = 2$	0.2120 (0.0088)	0.0768 (0.0050)	0.0580 (0.0041)	0.0446 (0.0037)	0.0123 (0.0022)	0.0035 (0.0011)	0.0026 (0.0011)	0.5867 (0.0111)
$h(j) = 3$	0.1552 (0.0051)	0.0766 (0.0035)	0.0670 (0.0032)	0.0503 (0.0029)	0.0174 (0.0018)	0.0039 (0.0009)	0.0021 (0.0006)	0.6255 (0.0072)
$h(j) = 4$	0.2465 (0.0083)	0.1131 (0.0060)	0.0818 (0.0048)	0.0629 (0.0045)	0.0134 (0.0021)	0.0018 (0.0009)	0.0005 (0.0005)	0.4792 (0.0107)
Tightness moment							-0.0291 (0.0037)	

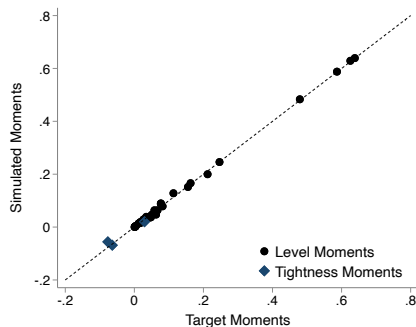
Moment Fitting



Moment Fitting Robustness



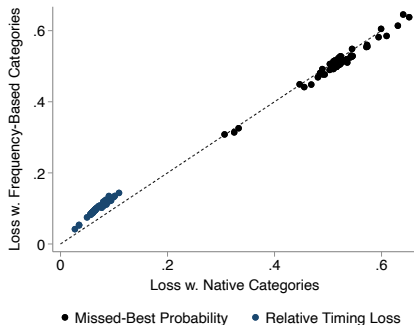
Alternative Moment Set A



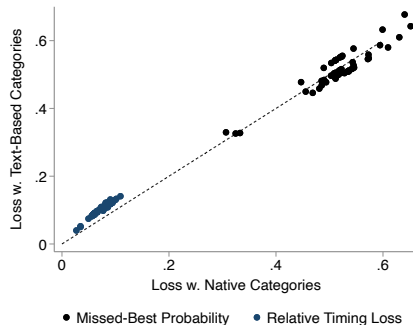
Alternative Moment Set B

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Headline Result Robustness: Screening-Value Costs



Frequency-Based Category Scores



Text-Based Category Scores

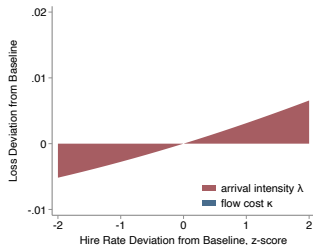
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Selected Counterfactuals on the Frontier

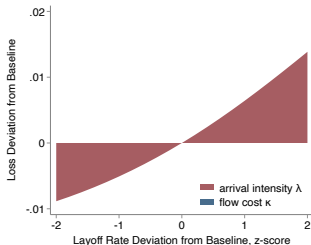
Counterfactual	Expected Initiation-Day Bin	$\bar{P}^{\text{exp}}(\hat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	1.213	0.4079	0.5156	0.0656
$\kappa_j \times 0.5$	1.326	0.3726	0.4937	0.0620
$\kappa_j \times 2$	1.044	0.4579	0.5482	0.0711
Mean $\log(\theta)$ to p_{10}	1.248	0.3754	0.5081	0.0643
Mean $\log(\theta)$ to p_{90}	1.182	0.4359	0.5221	0.0667

[▶ back](#)

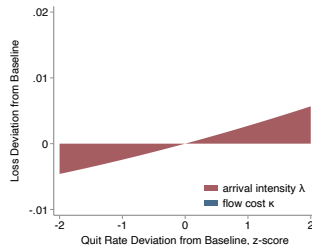
Quantifying Firms' Tradeoff: Counterfactuals



by Hires



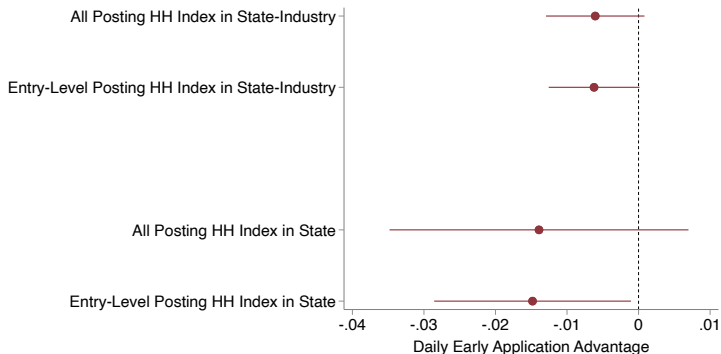
by Layoffs/Discharges



by Quits

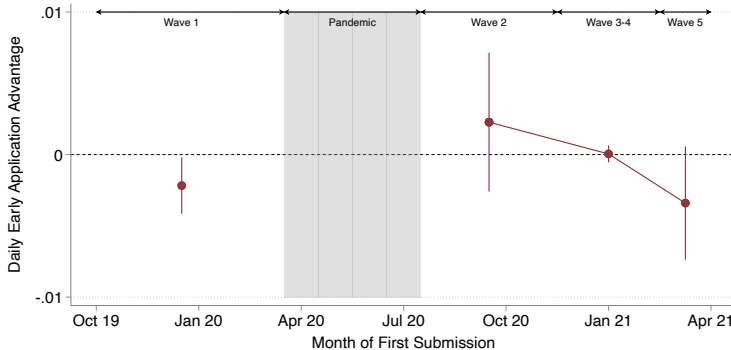
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External Validity: Market Concentration

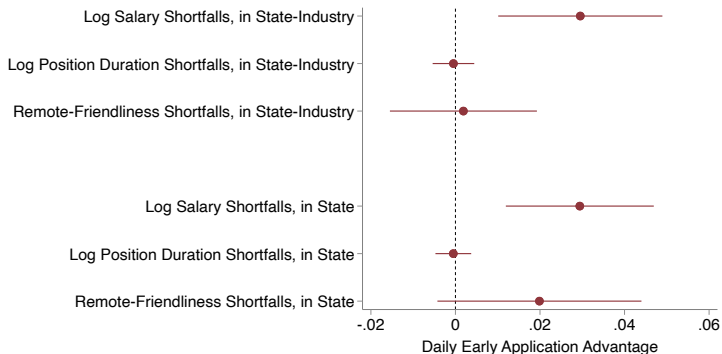


► welfare implications

External Validity: Local Recruitment Prominence

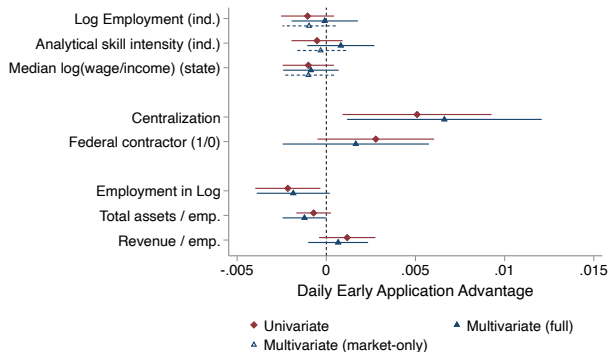
[► welfare implications](#)

External Validity: Offer Competitiveness



► welfare implications

External Validity: Market, Firm and Vacancy Characteristics



► early application advantage

► welfare implications

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